Cell-Type-Specific Biomarkers in Human Liver Fibrosis

Reproducible single-cell discovery pipeline, full report

2026-06-01

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1 Overview

This report is generated by the liver-fibrosis pipeline. It summarizes QC, integration, annotation, donor-aware differential expression, pathway and cell-cell communication mechanism analysis, and the AI/ML biomarker prioritization that yields the ranked candidate table. Method rationale and citations are in `docs/methods.md`; written answers to the screening questions are in `docs/written_answers.md`.

2 QC summary

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	sample	dataset	modality	n_cells_before	n_cells_after	n_genes_after	pct_kept	doul
0	GSM4041150	gse136103	scRNA	1863	1478	14696	79.3	4
1	GSM4041151	gse136103	scRNA	1039	655	15112	63.0	0
2	GSM4041152	gse136103	scRNA	478	299	14288	62.6	1
3	GSM4041153	gse136103	scRNA	6571	4762	15290	72.5	1056
4	GSM4041154	gse136103	scRNA	4688	4128	17643	88.1	1
5	GSM4041155	gse136103	scRNA	3295	2429	15511	73.7	39
6	GSM4041156	gse136103	scRNA	3230	2550	18383	78.9	4
7	GSM4041157	gse136103	scRNA	1156	820	15961	70.9	5
8	GSM4041158	gse136103	scRNA	4937	4123	15857	83.5	1
9	GSM4041159	gse136103	scRNA	3306	2826	17871	85.5	1
10	GSM4041160	gse136103	scRNA	5122	4044	15864	79.0	2
11	GSM4041161	gse136103	scRNA	2380	1764	16360	74.1	9
12	GSM4041162	gse136103	scRNA	1686	1231	17549	73.0	1
13	GSM4041163	gse136103	scRNA	2046	1231	17471	60.2	3
14	GSM4041164	gse136103	scRNA	3412	2698	17415	79.1	4
15	GSM4041165	gse136103	scRNA	3673	3031	18698	82.5	2
16	GSM4041166	gse136103	scRNA	1985	1568	16745	79.0	1
17	GSM4041167	gse136103	scRNA	3602	2955	17763	82.0	0
18	GSM4041168	gse136103	scRNA	4819	3801	16185	78.9	0
19	GSM4041169	gse136103	scRNA	2922	2061	15695	70.5	5

gse244832

	sample	dataset	modality	n_cells_before	n_cells_after	n_genes_after	pct_kept	doublets_
0	JB288	gse244832	snRNA	7492	5650	22420	75.4	87
1	JB289	gse244832	snRNA	3447	2152	17367	62.4	12
2	JB290	gse244832	snRNA	9148	6693	23584	73.2	10
3	JB303	gse244832	snRNA	8739	6764	21827	77.4	28
4	JB304	gse244832	snRNA	6531	3702	19845	56.7	35
5	JB305	gse244832	snRNA	7134	4408	21759	61.8	19
6	JB317	gse244832	snRNA	6223	4350	20922	69.9	5
7	JB318	gse244832	snRNA	6867	5158	22041	75.1	2
8	JB319	gse244832	snRNA	7541	5447	23001	72.2	4
9	JB320	gse244832	snRNA	8056	5793	21917	71.9	3

	sample	dataset	modality	n_cells_before	n_cells_after	n_genes_after	pct_kept	doublets_r
10	JB321	gse244832	snRNA	7325	5432	22550	74.2	0
11	JB322	gse244832	snRNA	8703	6863	22075	78.9	5
12	JB336	gse244832	snRNA	4260	3360	20460	78.9	23
13	JB337	gse244832	snRNA	7960	6320	22100	79.4	5
14	JB338	gse244832	snRNA	7244	5412	21832	74.7	2
15	JB339	gse244832	snRNA	6936	5308	21504	76.5	10
16	JB340	gse244832	snRNA	7452	5835	22157	78.3	6
17	JB341	gse244832	snRNA	7165	5327	21823	74.3	1

3 Integration (scIB) and annotation

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	n_cells	n_batches	batch_silhouette	condition_silhouette	condition_knn_accuracy	fibrosis_signal
0	48454	20	-0.144496	0.021229	0.641308	True

	leiden	n_cells	cell_type	compartment	score_Hepatocyte	score_Cholangiocyte
0	0	7302	T_NK	lymphoid	-0.039	-0.041
1	1	6661	T_NK	lymphoid	-0.040	-0.060
2	2	6547	T_NK	lymphoid	-0.048	-0.046
3	3	5056	Monocyte_MoMac	macrophage_monocyte	0.105	-0.047
4	4	4566	T_NK	lymphoid	-0.051	-0.046
5	5	3838	T_NK	lymphoid	-0.041	-0.025
6	6	3314	Endothelial_vasc	endothelial	-0.059	-0.088
7	7	1969	Endothelial_vasc	endothelial	-0.051	-0.068
8	8	1911	HSC_mesenchyme	stellate_myofibroblast	-0.006	-0.031
9	9	1758	Kupffer_resident	macrophage_monocyte	-0.009	-0.056
10	10	1462	Cholangiocyte	cholangiocyte	0.710	1.676
11	11	1350	LSEC	endothelial	-0.031	-0.119
12	12	1220	B_plasma	lymphoid	-0.029	-0.050
13	13	886	T_NK	lymphoid	-0.055	-0.037
14	14	237	Endothelial_vasc	endothelial	-0.020	-0.072
15	15	229	DC	myeloid_other	-0.075	-0.088
16	16	148	B_plasma	lymphoid	-0.004	-0.033

gse244832

	n_cells	n_batches	batch_silhouette	condition_silhouette	condition_knn_accuracy	fibrosis_signal
0	93974	18	-0.176235	-0.052978	0.546396	False

	leiden	n_cells	cell_type	compartment	score_Hepatocyte	score_Cholangiocyte
0	0	15961	Hepatocyte	hepatocyte	0.742	-0.012
1	1	15528	Hepatocyte	hepatocyte	0.937	-0.011
2	2	12761	Hepatocyte	hepatocyte	0.912	-0.012
3	3	12270	Hepatocyte	hepatocyte	0.722	0.022
4	4	8483	LSEC	endothelial	0.051	-0.034
5	5	7028	Hepatocyte	hepatocyte	1.077	-0.014
6	6	6761	LSEC	endothelial	-0.247	-0.046
7	7	6492	HSC_mesenchyme	stellate_myofibroblast	0.138	-0.014
8	8	3364	Kupffer_resident	macrophage_monocyte	0.304	-0.022
9	9	1607	Hepatocyte	hepatocyte	0.755	-0.023
10	10	1309	HSC_mesenchyme	stellate_myofibroblast	0.062	-0.016
11	11	861	Hepatocyte	hepatocyte	0.822	0.008
12	12	754	Cholangiocyte	cholangiocyte	0.058	0.666
13	13	263	Endothelial_vasc	endothelial	-0.219	-0.043
14	14	219	Hepatocyte	hepatocyte	1.091	-0.031
15	15	155	Hepatocyte	hepatocyte	0.221	-0.044
16	16	98	Hepatocyte	hepatocyte	0.373	0.091
17	17	60	Hepatocyte	hepatocyte	0.568	-0.062

4 Ranked biomarker / target candidates

The headline list is gated on differential-expression significance (compartment-level DESeq2 *or* a disease-associated-subcluster marker) and on cell-type specificity, so each candidate is a genuine, compartment-specific disease signal rather than an abundant lineage marker. Candidates flagged `in_disease_niche` are recovered from a disease-associated subpopulation that compartment-level pseudobulk averages out (e.g. scar-associated endothelium / macrophages).

	rank	gene	compartment	composite	log2FoldChange	padj	de_significant
0	1.0	PDGFRA	stellate_myofibroblast	0.671195	3.607818	7.830948e-10	True
1	2.0	ACKR1	endothelial	0.634854	4.058903	1.200014e-07	True
2	3.0	LUM	stellate_myofibroblast	0.624039	2.771089	1.927909e-11	True
3	4.0	DCN	stellate_myofibroblast	0.611797	2.094342	4.732443e-06	True
4	5.0	TREM2	macrophage_monocyte	0.606202	0.693682	2.765572e-02	True

	rank	gene	compartment	composite	log2FoldChange	padj	de_significan
5	6.0	PLVAP	endothelial	0.580863	0.239088	3.761008e-01	False
7	7.0	COL3A1	stellate_myofibroblast	0.552956	3.555308	9.999196e-16	True
9	8.0	MMP7	cholangiocyte	0.545108	1.956347	1.235578e-04	True
10	9.0	SPP2	cholangiocyte	0.543718	4.314999	8.929990e-04	True
11	10.0	MMP2	stellate_myofibroblast	0.542169	3.328071	2.200474e-09	True
12	11.0	COL1A1	stellate_myofibroblast	0.537588	4.200858	3.664619e-31	True
15	12.0	RGCC	endothelial	0.511566	0.350916	2.877778e-01	False
16	13.0	COL1A2	stellate_myofibroblast	0.505817	3.246062	1.289774e-12	True
17	14.0	RGS4	cholangiocyte	0.494848	4.155586	8.929990e-04	True
22	15.0	VWF	endothelial	0.477451	2.781912	4.133218e-15	True
33	16.0	FABP4	endothelial	0.431836	4.128500	1.972624e-06	True
34	17.0	RGS10	macrophage_monocyte	0.428405	0.404238	2.762472e-02	False
49	18.0	IER3	macrophage_monocyte	0.405677	0.939377	8.366588e-05	True
78	19.0	PLXDC2	macrophage_monocyte	0.390093	0.314666	1.058874e-01	False
118	20.0	CTSH	macrophage_monocyte	0.362371	0.770582	6.240810e-04	True

5 Known-positive recall (sanity check)

Does the pipeline recover biomarkers already established in the human-liver-fibrosis literature, and where a known positive is missed, *why* (non-significant / filtered / absent)? Panel: `resources/known_positive_markers.tsv`.

	metric	value
0	n_known_positives	37.00
1	recall_selected	10.00
2	recall_selected_frac	0.27
3	surfaced_as_candidate	25.00
4	detected_significant_in_primary	26.00
5	rescued_by_validation_only	1.00

	gene	compartment	found_status	rank	primary_padj	validation_sta
0	COL1A1	stellate_myofibroblast	selected	11.0	3.664619e-31	val_tested_no
1	COL1A2	stellate_myofibroblast	selected	13.0	1.289774e-12	val_tested_no
2	COL3A1	stellate_myofibroblast	selected	7.0	9.999196e-16	val_tested_no
3	COL6A3	stellate_myofibroblast	DE_tested_nonsignificant	NaN	2.988873e-01	val_tested_no
4	COL5A1	stellate_myofibroblast	scored_not_selected	NaN	3.243491e-08	val_tested_no
5	ACTA2	stellate_myofibroblast	DE_tested_nonsignificant	NaN	4.781326e-01	val_tested_no

	gene	compartment	found_status	rank	primary_padj	validation_sta
6	TAGLN	stellate_myofibroblast	DE_tested_nonsignificant	NaN	5.233530e-01	val_tested_no
7	TIMP1	stellate_myofibroblast	scored_not_selected	NaN	1.927909e-11	val_tested_no
8	PDGFRB	stellate_myofibroblast	DE_tested_nonsignificant	NaN	9.113876e-01	val_tested_no
9	PDGFRA	stellate_myofibroblast	selected	1.0	7.830948e-10	val_tested_no
10	PDGFD	stellate_myofibroblast	scored_not_selected	NaN	2.669875e-03	validated
11	LOXL1	stellate_myofibroblast	scored_not_selected	NaN	1.476092e-03	val_tested_no
12	LOXL2	stellate_myofibroblast	scored_not_selected	NaN	7.423372e-03	val_tested_no
13	TGFB1	stellate_myofibroblast	scored_not_selected	NaN	5.980219e-03	val_tested_no
14	VCAN	stellate_myofibroblast	scored_not_selected	NaN	1.477132e-12	val_tested_no
15	DCN	stellate_myofibroblast	selected	4.0	4.732443e-06	val_tested_no
16	LUM	stellate_myofibroblast	selected	3.0	1.927909e-11	val_tested_no
17	SPARC	stellate_myofibroblast	scored_not_selected	NaN	3.742002e-02	val_tested_no
18	CCN2	stellate_myofibroblast	absent_primary	NaN	NaN	val_absent
19	FN1	stellate_myofibroblast	scored_not_selected	NaN	1.849887e-03	val_tested_no
20	SMOC2	stellate_myofibroblast	DE_tested_nonsignificant	NaN	8.753296e-01	validated
21	THY1	stellate_myofibroblast	scored_not_selected	NaN	9.999196e-16	val_tested_no
22	MMP2	stellate_myofibroblast	selected	10.0	2.200474e-09	val_tested_no
23	TREM2	macrophage_monocyte	selected	5.0	2.765572e-02	val_absent
24	CD9	macrophage_monocyte	scored_not_selected	NaN	2.941762e-03	val_tested_no
25	SPP1	macrophage_monocyte	scored_not_selected	NaN	1.362951e-01	val_tested_no
26	GPNMB	macrophage_monocyte	DE_tested_nonsignificant	NaN	8.999708e-01	val_tested_no
27	FABP5	macrophage_monocyte	DE_tested_nonsignificant	NaN	8.858496e-01	val_tested_no
28	CD63	macrophage_monocyte	DE_tested_nonsignificant	NaN	9.924401e-01	val_tested_no
29	LGMN	macrophage_monocyte	DE_significant_wrong_dir	NaN	2.087164e-02	val_tested_no
30	PLVAP	endothelial	selected	6.0	3.761008e-01	val_tested_no
31	ACKR1	endothelial	selected	2.0	1.200014e-07	val_absent
32	VWA1	endothelial	scored_not_selected	NaN	1.471575e-01	val_tested_no
33	CD34	endothelial	DE_tested_nonsignificant	NaN	9.528367e-01	val_tested_no
34	KRT19	cholangiocyte	scored_not_selected	NaN	9.376427e-01	val_tested_no
35	KRT7	cholangiocyte	scored_not_selected	NaN	4.204185e-01	val_tested_no
36	SOX9	cholangiocyte	DE_tested_nonsignificant	NaN	9.971689e-01	val_tested_no

6 Niche (subpopulation) structure

Disease-associated subclusters within each compartment (donor-aware enrichment, valid within a sorted compartment); their one-vs-rest markers are the subset biomarkers folded into the ranked list above — recovering signals (PLVAP, TREM2, …) that compartment-level pseudobulk buries.

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	compartment	subcluster	n_cells	frac_test	frac_ref	disease_log2fc	dise
9	stellate_myofibroblast	stellate_myofibroblast:8	50	0.1089	0.0035	4.613	Tru
44	lymphoid	lymphoid:10	95	0.0072	0.0003	2.710	Tru
23	endothelial	endothelial:10	236	0.0627	0.0124	2.249	Tru
32	endothelial	endothelial:9	291	0.0539	0.0116	2.119	Tru
27	endothelial	endothelial:4	647	0.1040	0.0269	1.911	Tru
21	endothelial	endothelial:0	1093	0.2446	0.0660	1.874	Tru
22	endothelial	endothelial:1	975	0.2054	0.0764	1.416	Tru
16	macrophage_monocyte	macrophage_monocyte:5	550	0.1167	0.0459	1.328	Tru
1	stellate_myofibroblast	stellate_myofibroblast:1	258	0.5364	0.2283	1.229	Tru
50	lymphoid	lymphoid:7	1130	0.0575	0.0258	1.124	Tru
37	cholangiocyte	cholangiocyte:4	191	0.1687	0.0799	1.069	Tru
31	endothelial	endothelial:8	305	0.0505	0.0255	0.958	Tru
8	stellate_myofibroblast	stellate_myofibroblast:7	95	0.1592	0.0892	0.829	Tru
15	macrophage_monocyte	macrophage_monocyte:4	753	0.1478	0.0835	0.816	Tru
34	cholangiocyte	cholangiocyte:1	246	0.2533	0.1565	0.691	Tru
46	lymphoid	lymphoid:3	4756	0.2080	0.1343	0.628	Tru

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	compartment	subcluster	n_cells	frac_test	frac_ref	disease_log2fc	dise
8	stellate_myofibroblast	stellate_myofibroblast:8	220	0.4255	0.0000	8.737	Tru
38	hepatocyte	hepatocyte:8	219	0.0711	0.0007	5.372	Tru
20	endothelial	endothelial:10	266	0.0685	0.0037	3.890	Tru
29	endothelial	endothelial:9	437	0.0465	0.0048	3.023	Tru
36	hepatocyte	hepatocyte:6	1171	0.0331	0.0044	2.660	Tru
3	stellate_myofibroblast	stellate_myofibroblast:3	1085	0.1218	0.0507	1.249	Tru
4	stellate_myofibroblast	stellate_myofibroblast:4	943	0.1233	0.0535	1.189	Tru
16	macrophage_monocyte	macrophage_monocyte:6	135	0.0544	0.0246	1.111	Tru
28	endothelial	endothelial:8	470	0.0348	0.0209	0.707	Tru

7 Cross-dataset reproducibility

	gene	compartment	primary_lfc	primary_padj	valid_lfc	valid_padj	in_validati
0	COL1A1	stellate_myofibroblast	4.200858	3.664619e-31	0.001829	0.997055	True
1	COL3A1	stellate_myofibroblast	3.555308	9.999196e-16	0.018237	0.929306	True
2	RARRES1	stellate_myofibroblast	5.189493	9.999196e-16	0.014734	0.924622	True
3	THY1	stellate_myofibroblast	3.737300	9.999196e-16	0.015182	0.850957	True
4	SPON2	stellate_myofibroblast	3.677740	5.609830e-15	0.017549	0.948677	True
5	MMP23B	stellate_myofibroblast	3.902443	9.109332e-15	0.019165	0.884926	True
6	SERPINF1	stellate_myofibroblast	3.986592	5.871991e-14	0.101618	0.471905	True
7	COL1A2	stellate_myofibroblast	3.246062	1.289774e-12	0.038016	0.812575	True
8	VCAN	stellate_myofibroblast	5.709139	1.477132e-12	0.182225	0.319615	True
9	LUM	stellate_myofibroblast	2.771089	1.927909e-11	0.058411	0.565770	True
10	PCOLCE	stellate_myofibroblast	3.178234	1.927909e-11	-0.012877	0.946557	True
11	TIMP1	stellate_myofibroblast	3.679754	1.927909e-11	0.033817	0.749899	True
12	IGF1	stellate_myofibroblast	5.306151	2.540315e-11	0.014627	0.872871	True
13	F2R	stellate_myofibroblast	3.704220	6.420909e-10	0.071603	0.550206	True
14	PTGIS	stellate_myofibroblast	4.439223	7.728581e-10	0.016787	0.918349	True
15	PDGFRA	stellate_myofibroblast	3.607818	7.830948e-10	0.653297	0.054569	True
16	FBLN1	stellate_myofibroblast	3.314160	1.177560e-09	0.008589	0.965974	True
17	MEG3	stellate_myofibroblast	2.529856	1.177560e-09	-0.014807	0.951102	True
18	EFEMP1	stellate_myofibroblast	3.329391	1.189040e-09	1.743685	0.005329	True
19	MMP2	stellate_myofibroblast	3.328071	2.200474e-09	0.037307	0.740484	True

8 Mechanism: pathways and cell-cell communication

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	compartment	condition	source	mean_activity	level
0	cholangiocyte	cirrhotic	Androgen	1.0598	pathway_progeny
1	cholangiocyte	healthy	Androgen	0.8873	pathway_progeny
2	endothelial	cirrhotic	Androgen	0.4854	pathway_progeny
3	endothelial	healthy	Androgen	1.2176	pathway_progeny
4	lymphoid	cirrhotic	Androgen	0.3709	pathway_progeny
5	lymphoid	healthy	Androgen	0.3829	pathway_progeny
6	macrophage_monocyte	cirrhotic	Androgen	0.5556	pathway_progeny
7	macrophage_monocyte	healthy	Androgen	0.6782	pathway_progeny
8	myeloid_other	cirrhotic	Androgen	0.6113	pathway_progeny
9	myeloid_other	healthy	Androgen	0.6975	pathway_progeny
10	stellate_myofibroblast	cirrhotic	Androgen	0.2392	pathway_progeny
11	stellate_myofibroblast	healthy	Androgen	0.0296	pathway_progeny

	compartment	condition	source	mean_activity	level
12	cholangiocyte	cirrhotic	EGFR	2.2357	pathway_progeny
13	cholangiocyte	healthy	EGFR	1.9298	pathway_progeny
14	endothelial	cirrhotic	EGFR	2.4658	pathway_progeny

	source	target	ligand_complex	receptor_complex	fibrotic_conditio
0	myeloid_other	cholangiocyte	SCT	SCTR	cirrhotic
1	cholangiocyte	lymphoid	DEFB1	CCR6	cirrhotic
2	cholangiocyte	endothelial	CXCL6	ACKR1	cirrhotic
3	stellate_myofibroblast	endothelial	COL1A1	ITGA10_ITGB1	cirrhotic
4	stellate_myofibroblast	endothelial	COL1A1	FLT4	cirrhotic
5	stellate_myofibroblast	endothelial	COL1A1	ITGA5	cirrhotic
6	stellate_myofibroblast	endothelial	COL3A1	ITGA10_ITGB1	cirrhotic
7	stellate_myofibroblast	endothelial	COL1A1	CD93	cirrhotic
8	stellate_myofibroblast	cholangiocyte	COL1A1	DDR1	cirrhotic
9	stellate_myofibroblast	stellate_myofibroblast	COL1A1	ITGA1_ITGB1	cirrhotic
10	stellate_myofibroblast	stellate_myofibroblast	COL1A1	DDR2	cirrhotic
11	stellate_myofibroblast	cholangiocyte	COL1A1	ITGAV_ITGB8	cirrhotic
12	stellate_myofibroblast	stellate_myofibroblast	COL1A2	ITGA1_ITGB1	cirrhotic
13	stellate_myofibroblast	stellate_myofibroblast	THBS2	NOTCH3	cirrhotic
14	endothelial	endothelial	CCL14	ACKR1	cirrhotic

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### gse244832
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	compartment	condition	source	mean_activity	level
0	cholangiocyte	MASH	Androgen	3.1403	pathway_progeny
1	cholangiocyte	MASL	Androgen	2.6690	pathway_progeny
2	cholangiocyte	normal	Androgen	1.8914	pathway_progeny
3	endothelial	MASH	Androgen	2.1593	pathway_progeny
4	endothelial	MASL	Androgen	2.2036	pathway_progeny
5	endothelial	normal	Androgen	2.4728	pathway_progeny
6	hepatocyte	MASH	Androgen	4.8672	pathway_progeny
7	hepatocyte	MASL	Androgen	4.7303	pathway_progeny
8	hepatocyte	normal	Androgen	4.5555	pathway_progeny
9	macrophage_monocyte	MASH	Androgen	2.2043	pathway_progeny
10	macrophage_monocyte	MASL	Androgen	1.8532	pathway_progeny
11	macrophage_monocyte	normal	Androgen	1.6571	pathway_progeny
12	stellate_myofibroblast	MASH	Androgen	2.5075	pathway_progeny

	compartment	condition	source	mean_activity	level
13	stellate_myofibroblast	MASL	Androgen	2.4753	pathway_progeny
14	stellate_myofibroblast	normal	Androgen	2.4165	pathway_progeny

	source	target	ligand_complex	receptor_complex	fibrotic_conditio
0	cholangiocyte	cholangiocyte	EFNA5	EPHA6	MASH
1	stellate_myofibroblast	cholangiocyte	LAMA2	ITGA2_ITGB1	MASH
2	endothelial	cholangiocyte	FGF23	FGFR2	MASH
3	stellate_myofibroblast	cholangiocyte	FGF7	FGFR2	MASH
4	stellate_myofibroblast	cholangiocyte	LAMA2	ITGA6_ITGB4	MASH
5	cholangiocyte	stellate_myofibroblast	EFNA5	EPHA3	MASH
6	endothelial	cholangiocyte	EFNB2	EPHA6	MASH
7	cholangiocyte	cholangiocyte	EFNA5	EPHB6	MASH
8	cholangiocyte	cholangiocyte	LAMC2	ITGAV_ITGB8	MASH
9	stellate_myofibroblast	cholangiocyte	COL3A1	ITGA2_ITGB1	MASH
10	stellate_myofibroblast	cholangiocyte	LAMC3	ITGA2_ITGB1	MASH
11	endothelial	cholangiocyte	HSPG2	ITGA2	MASH
12	stellate_myofibroblast	cholangiocyte	COL6A3	ITGA2_ITGB1	MASH
13	stellate_myofibroblast	cholangiocyte	COL4A1	ITGA2_ITGB1	MASH
14	cholangiocyte	stellate_myofibroblast	NCAM1	CACNA1C	MASH